

Open Source Tools For EMBEDDED DEVELOPERS

The embedded community has well embraced open source tools for design and development, but is open source fit for all embedded projects? Open source tools like gcc, gdb and Eclipse have gained popularity, but there are still certain issues that need to be resolved

■ EFY BUREAU

Open source software has captured the attention of many, not because it's free, but because it gives freedom! For embedded design, where you need freedom to experiment and innovate, open source has become hot-bed of opportunities. With open source, at least you save on the licence fee, and if you are smart enough, you will also save the support fee.

"The open source belief is, 'If you have the itch, you scratch it,'" thoughtfully shares R.M. Nair, vice president at Acmet Technologies. But in certain projects, maturity of open source tools may jeopardise the development process. Open source pitch may not take the googly in the case of certain hardware components in the game of embedded systems.

"The preference of using open source tools over proprietary tools is primarily based on the costing reasons. But, for all reasons, we know for sure that proprietary tools are better in terms of performance in the case of specific targets for a particular operation," shares P.S. Walia, project manager, Dexcel.

Choice of a tool or a tool chain depends on many parameters like the nature of projects that your organisation works on, cost investments and in-house engineering expertise. "Compilers are the most preferred open source tools. However, in case propri-



Eclipse IDE

etary tools are available that combine many functions like compilation and debugging, they are preferred. For example, gcc port for MSP430 is available, which provides debugging via gdb. However, Texas Instruments provides code composer MCU edition and IAR provides a complete IDE and embedded workbench, which contain both a compiler and an effective debugger," says Praval Jain, vice president-networks, Virtualwire Technologies.

Jain adds, "If the embedded system is running Linux, open-source tools are the only choice in any case."

Open source benefits

"Open source, by definition, gives you the freedom and flexibility to use the tool. So it provides a very good foundation to jumpstart development activities," says Dhaval Vasa, project champion-embedded division, elfochips. "With the licensing of open source, the projects can use other open source projects as off-the-shelf components." This speeds up the development process.

Another benefit of open source is that it makes software portable. "There are several open source tools available for a wide range of microcontrollers.